



MEETING ROOM RESERVATION SYSTEM

PRESENTED BY: MARY WARUGURU WANJIRU

REG. NO.: 20/05102

SUPERVISOR: JACOB OLOO

**SYSTEM REQUIREMENT SPECIFICATION PRESENTED TO THE SCHOOL OF
TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF A BACHELOR'S DEGREE IN INFORMATION AND COMMUNICATION
TECHNOLOGY AT KCA UNIVERSITY**

SUBMISSION DATE JANUARY 2025

ABSTRACT

This project proposes the development of a web-based Meeting Room Reservation Management System for Kenya Law to replace the existing manual process. The current system, reliant on verbal communication with a secretary, results in inefficiencies such as scheduling conflicts, lack of transparency, and delays. The proposed system will enhance efficiency, transparency, and accessibility by allowing employees to book rooms independently. It will include real-time availability tracking, automated notifications, and integration with Google Calendar. The system will follow the Rapid Application Development (RAD) model, ensuring iterative improvement based on user feedback. Testing and validation will ensure functionality, with a projected completion timeline of 10 weeks.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Jacob Oloo, for his invaluable guidance and support throughout this project. I also extend my appreciation to my family, friends, and colleagues who provided encouragement and assistance during this research

DECLARATION

I, Mary Waruguru Wanjiru, declare that this project is my original work and has never been submitted to this or any other university for academic credit.

Supervisor **Signature:** _____ **Date:** _____

Student **Signature:** _____ **Date:** _____

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1. Introduction

1.1 Purpose

This document outlines the software requirements for the **Meeting Room Reservation System (MRRS)**, a web-based application designed to facilitate the reservation and management of meeting rooms within an organization. The system will allow users to book rooms, view availability, and manage reservations efficiently. This SRS covers the entire system and is intended to guide the development team in building the application using Java as the primary programming language.

1.2 Intended Audience

This document is intended for the following audiences:

- **Developers:** To understand the functional and non-functional requirements for implementation.
- **Project Managers:** To oversee the development process and ensure alignment with business goals.
- **Testers:** To create test cases and validate the system against the requirements.
- **End Users:** To gain an understanding of the system's capabilities and features.

It is recommended that readers begin with the **Introduction** and **Overall Description** sections to gain a high-level understanding of the system.

Developers should focus on the **Functional Requirements** and **External Interface Requirements** sections, while testers should pay close attention to the **Nonfunctional Requirements** section.

1.3 Project Scope

The Meeting Room Reservation System aims to streamline the process of booking and managing meeting rooms within an organization. The system will provide the following key functionalities:

- User authentication and authorization.
- Real-time room availability tracking.
- Reservation creation, modification, and cancellation.
- Notifications and reminders for upcoming reservations.
- Reporting and analytics for room usage.

2. Overall Description

2.1 User Classes and Characteristics

The system will cater to the following user classes:

1. **Administrators:** Responsible for managing rooms, user accounts, and system settings.
2. **Company's Secretary /Head of Departments-** users who can book, modify, and cancel reservations.
3. **Employees:** users with limited access to book rooms, subject to approval.

Each user class will have distinct privileges and access levels.

2.2 Operating Environment

The system will operate in the following environment:

- **Hardware:** Standard servers with sufficient memory and processing power.
- **Operating System:** Linux/Windows.
- **Software:** Java Runtime Environment (JRE), Apache Tomcat, MySQL Database.
- **Browser Compatibility:** Chrome, Firefox, Edge (latest versions).

2.3 Design and Implementation Constraints

- The system must be developed using the MERN stack this (Mongo Data base, Express.js, React, Node.js)
- Should integrate with third-party calendar systems.
- The system must comply with organizational security policies.
- The user interface must adhere to the organization's branding guidelines.

2.4 User Documentation

The following user documentation will be provided:

- User Manual (PDF and online help).
- Quick Start Guide.
- Online user manuals.

- FAQs and troubleshooting guides.

2.5 Assumptions and Dependencies

- The system assumes that users have basic computer literacy.
- The system depends on the organization's existing network infrastructure.
- Third-party libraries (e.g., Spring Security, Hibernate), notification services (e.g., email/ will be used for development.

3. Functional Requirements

3.1 System Features

3.1.1 User Authentication and Authorization

- **Description:** Users must log in to access the system. Different roles have different access levels.
- **Priority:** High.
- **Stimulus/Response Sequences:**
 - User enters credentials → System validates and grants access.
 - Invalid credentials → System displays error message.

3.1.2 Room Reservation

- **Description:** Users can view room availability and book rooms.
- **Priority:** High.
- **Stimulus/Response Sequences:**
 - User selects date and time → System displays available rooms.
 - User selects a room → System confirms reservation.

3.1.3 Reservation Management

- **Description:** Users can modify or cancel existing reservations.
- **Priority:** Medium.
- **Stimulus/Response Sequences:**
 - User selects reservation → System allows modification or cancellation.

3.1.4 Notifications

- **Description:** Users receive email notifications for upcoming reservations.
- **Priority:** Medium.
- **Stimulus/Response Sequences:**
 - Reservation is created → System sends confirmation email.

4. External Interface Requirements

4.1 User Interfaces

The system will have a web-based interface with the following components:

- Login page.
- Intuitive Dashboard for room selection and booking and checking availability.
- Reservation form.
- Admin panel for managing rooms and users.
- Mobile-responsive design.

4.2 Hardware Interfaces

The system will run on standard servers with no special hardware requirements and Supports desktops, tablets, and smartphones.

4.3 Software Interfaces

The system will integrate with:

- Mongo database for data storage.
- SMTP server for email notifications.
- Integrates with Google Calendar

4.4 Communications Interfaces

The system will use HTTPS for web communication and SMTP server Email notifications for booking confirmations.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system must support up to 100 concurrent users without performance degradation

5.2 Safety Requirements

The system must ensure data integrity and prevent unauthorized access.

5.3 Security Requirements

- User passwords must be encrypted.
- The system must comply with GDPR for data protection.

5.4 Software Quality Attributes

- **Usability:** The system should be intuitive and easy to use.
- **Maintainability:** Code should be well-documented and modular.
- **Scalability:** The system should handle future growth in users and rooms.

6. Other Requirements

- The database must be backed up daily.

7. References

Spring Boot Documentation: <https://spring.io/projects/spring-boot>

MongoDB Documentation <https://www.mongodb.com/docs/manual/>

React Official Docs <https://react.dev/learn>

Node.js API Documentation <https://nodejs.org/docs/latest/api/>

Martin, J. (1991). *Rapid application development*. Macmillan Publishing.

Sharma, R. (2020). *Mastering Spring Boot 2.0: Build modern, cloud-native, and distributed systems using Java and Spring*. Packt Publishing.

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Appendix A: Glossary

- **MRRS:** Meeting Room Reservation System.
- **GDPR:** General Data Protection Regulation.
- **SMTP:** Simple Mail Transfer Protocol.
- **MERN:** MongoDB, Express.js, React, Node.js
- **JWT:** JSON Web Token
- **API:** Application Programming Interface